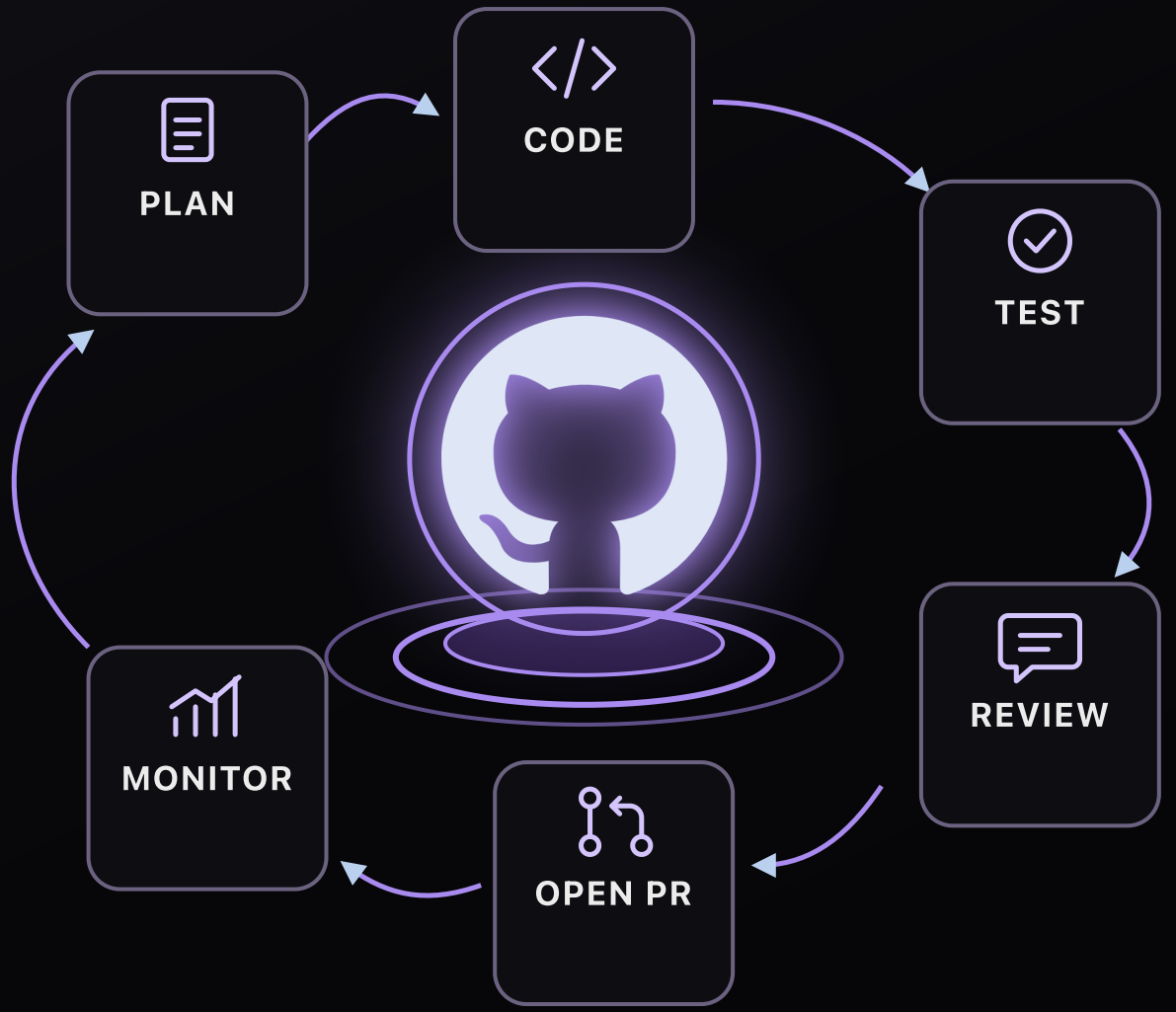
GitHub

GitHub Agentic Workflows

FROM AI THAT SUGGESTS TO AI THAT DOES

When did a page of documentation last mislead you?



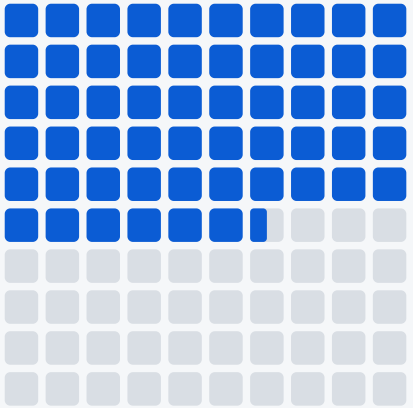
— 01 / THE PROBLEM

Outdated documentation already costs developers time

56.5%

of 12,374 developers say **outdated information** makes it hard to get the context they need.

In plain words: more than half



6,989 named it
56.5% of 12,374

5,385 did not
derived: 12,374 minus 6,989

1 square is about 1% of respondents

Industry survey (multi-select, about context in general), not our repositories · our own drift is not yet measured.

— 01B / THE PROBLEM

My earlier attempt solved part of the problem, and left a gap

What I built

tool name | what it did

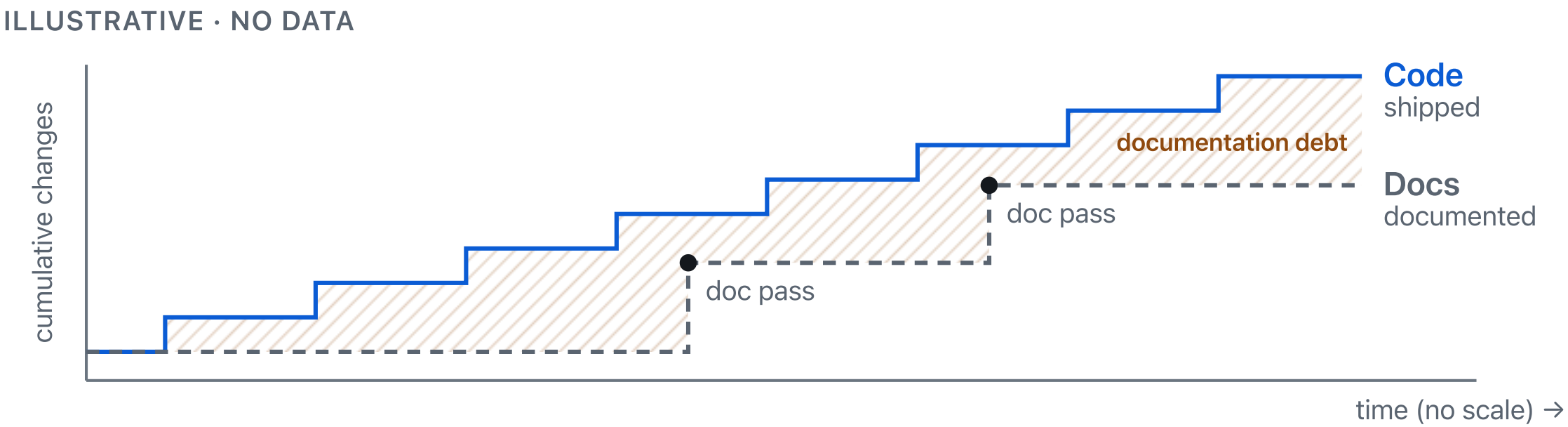
What it left undone

the remaining gap, one concrete instance

> Still blank: my own story. Nothing here is a claim until I supply it. Keep it to one tool, one limitation, one moment.

— 02 / THE MECHANISM

Documentation can fall behind code between doc passes



> Every shipped change adds to it; each doc pass pays only some back.

Fig. 1. Illustrative, no data, extends Cunningham's (1992) technical debt metaphor to documentation.

— 03 / THE ASK

The ask is four weeks, one repository, and docs only

4 weeks

then a stop-or-go decision

1 repository

not yet chosen

Docs only

by instruction, reviewed in the diff

Draft PR

a proposed change, not yet merged; a person decides

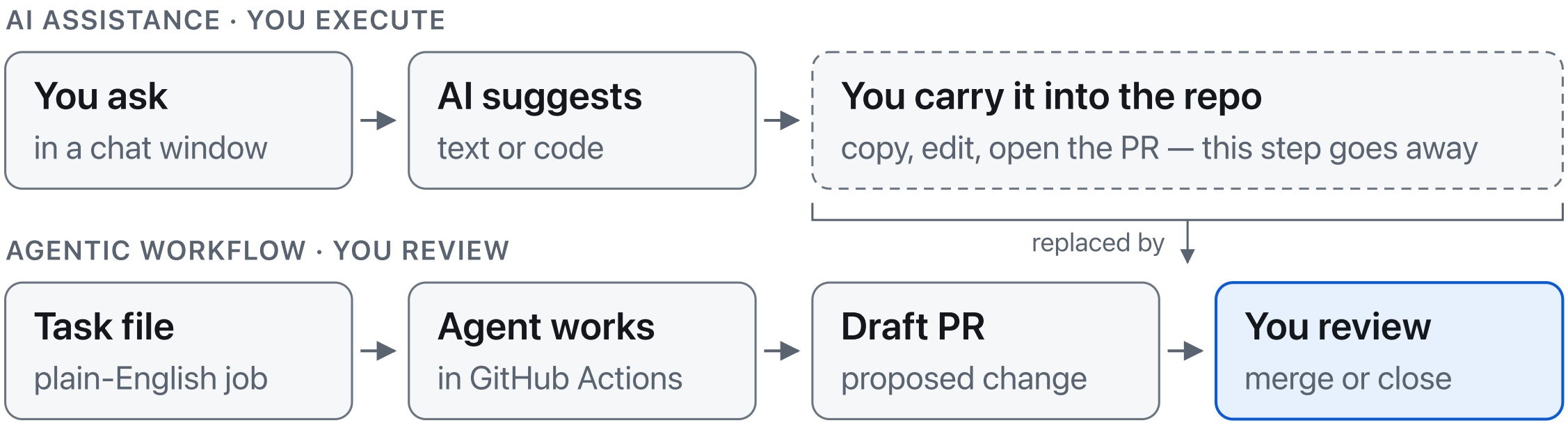
> Everything after this slide is evidence for or against that ask. The cost is unknown; the pilot measures it.

Still blank: **repository** **owner** **budget** **week-four thresholds**

This deck's proposal · public preview since 11 Jun 2026 · bounds: slide 14.

— 04 / THE SOLUTION

An agentic workflow hands the human a prepared change to review



> The agent drafts, a person decides what merges: the work moves from writing to judging.

Fig. 2 · Illustrative, this deck's framing · `draft: true` is enforced PR policy · reviewer minutes not yet measured · Actions is GitHub's built-in automation.

— 05 / WHY AN AGENT

Scripts check fixed rules; an agent is for the part that needs reading

First ask: can a fixed rule decide this?

Yes: use a script

Broken links · missing API entries · formatting

Cheaper, faster, predictable.

No, it needs reading: use an agent

“Does this paragraph still describe what the function does?”

Few fixed rules express that.

> Use the deterministic tool wherever one solves the task.

TODO · decide before week one: by default CI does not run on the agent's pull requests (workflow-token PRs do not trigger it); gh-aw documents an optional token that does. Framing is this deck's.

gh-aw docs: a third pillar beside CI and CD · Triggering CI

— 06 / THE REAL EXAMPLE

A documentation agent's real pull request adds two glossary entries in one file

A workflow named “[docs] Update glossary - daily scan” opened it. It adds two glossary entries.

ADDED · docs/…/reference/glossary.md · +8 −0 · TRIMMED

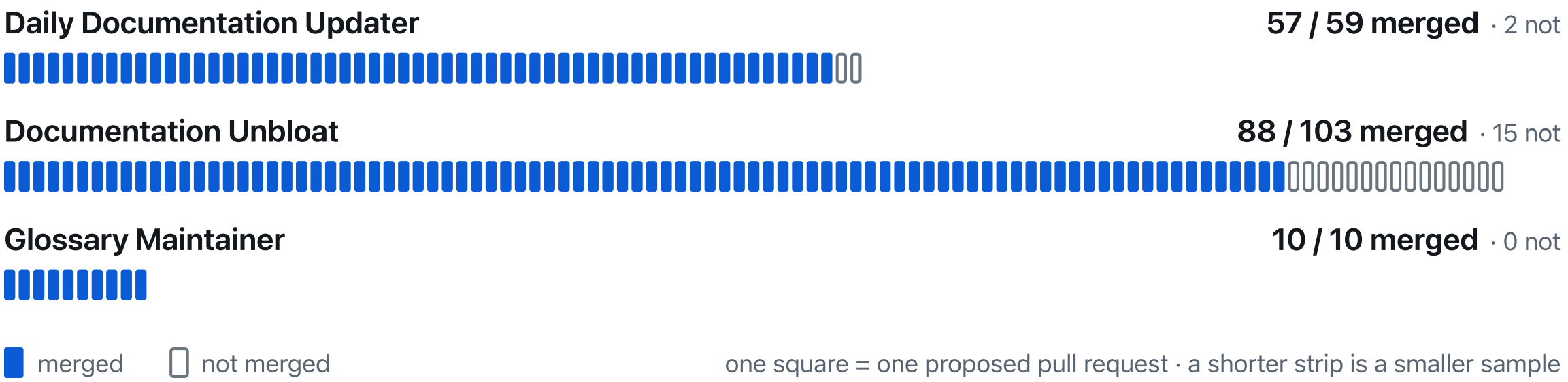
```
+ ### Work Queue (`tools.work-queue`)
+ A native, Git-backed queue for fair scheduling of
+ Work with immutable task dependency graphs. …
+
+ ### Claim (Work Queue)
+ Authorization for one attempt at a Work item in
+ a native work queue … See Work Queues.
```

> One example, not a rate: it shows the size of a change, not that an agent repairs stale docs.

Public example · github/gh-aw #68654, merged 14 Oct 2026 · GitHub's own repository, not our pilot · text trimmed with …

— 07 / THE EVIDENCE

In three of GitHub's documentation workflows, most pull requests were merged, but not all



> Selected samples from other people's repositories, not a forecast. The same post lists a lower one: 9 of 21 (43%).

Fig. 4 · Reported by GitHub, 13 Jan 2026, repository not stated, not verified by us · these were ready-for-review pull requests, not

— 08 / THE RISK

It will be wrong sometimes, and that has to be survivable

Confidently wrong

It matches a page to code it misread.

LIMIT
A person reads the diff.

Too wide a diff

A one-page fix touches twenty files.

LIMIT
Draft PR and patch-size limits; docs only by instruction.

Planted instructions

A pasted issue says “ignore your rules”.

LIMIT
Read-only agent job; checks before any write.

Spend, no merge

AI usage charges (credits) burn whether or not anything lands.

LIMIT
Spend caps: TODO

> Each has a limit we set; none is a guarantee, so a person still reads the diff.

Failure modes this deck names · prompt injection is documented by GitHub; the rest follow from the task.

TODO · our own error rate is unmeasured — week one's staged preview is the first measurement.

Threat detection · Cost management

The limits live in a file you can read, not in the model

Think of it as a new colleague's onboarding sheet: when they work, what they can touch, what they can hand in. The block on top of the file is its settings (“frontmatter”); the text below is the job description.

```
---
on:                                    1
  push:
    branches: [main]
permissions: read-all                  2
safe-outputs:                          3
  create-pull-request:
    draft: true
---
# Update Docs                          4
Ensure every code-level change is
mirrored by clear, accurate, and
stylistically consistent documentation.
```

1. **Shift** `on:`
   Every push to main
2. **Access badge** `permissions`
   Reads the repository
3. **Outbox** `safe-outputs`
   Draft pull requests
4. **Job description** Markdown body
   Plain-English instructions

Trimmed from githubnext/agentics update-docs.md · PR policy

The same file also lets the agent run commands and fetch pages, so the write limit has to hold

```
network: defaults          # 1
tools:                     # 2
  github: {toolsets: [all]}
  web-fetch:
  bash: true
safe-outputs:
  create-pull-request:
    draft: true
    protected-files: fallback-to-issue   # 3
timeout-minutes: 15        # 4
```

1 · `network: defaults`
An allowlist of default hosts, not “no network”

2 · `tools`
Bash inside the Actions VM, web fetch, all GitHub toolsets

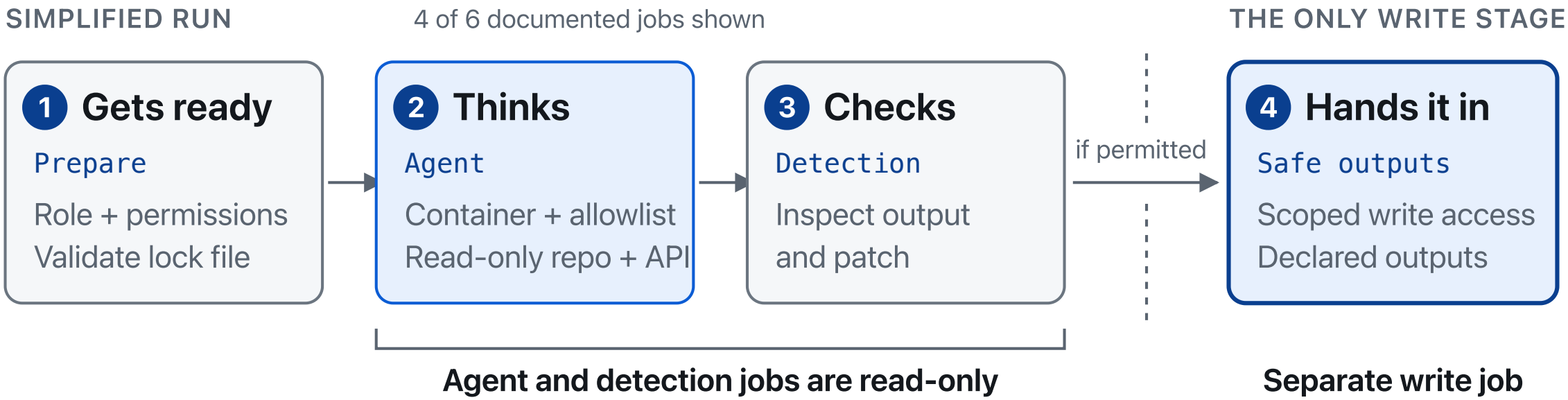
3 · `protected-files`
Routes changes to sensitive files to a review issue; it does not limit a PR to docs

4 · `timeout-minutes`
Stops a run after 15 minutes

> TODO · add `allowed-files` (a docs path) to the pilot copy if “docs only” is to be enforced, and confirm in a trial; pilot tool list is not chosen.

Real file, trimmed · githubnext/agentics workflows/update-docs.md

The agent job is read-only; a separate job does the writing



A separate step applies outputs; a draft PR and human merge are the backstop.

Fig. 5. Detection is a layer, not a guarantee: by default (`continue-on-error: true`) warnings give a caution, not a block; least privilege (Saltzer & Schroeder, 1975).

— 11 / COST

A cheap run nobody merges is still expensive

The right unit is cost per **accepted** change.

AI credits
from `gh aw logs`
NOT MEASURED

+

Actions minutes
from billing
NOT MEASURED

+

Reviewer time
from the pilot
NOT MEASURED

÷

Accepted changes
a reviewer said yes
NOT MEASURED

> Cost per accepted change: add the three, divide by what a reviewer accepted.

Fig. 6a · no run yet, so nothing here is measured · AIC = AI credit, 1 AIC ≈ $0.01 estimated · billing may differ.

TODO · Agree pilot limits; measure cost and review time · formula and symbols: Appendix 9.

Each week of the pilot has one job

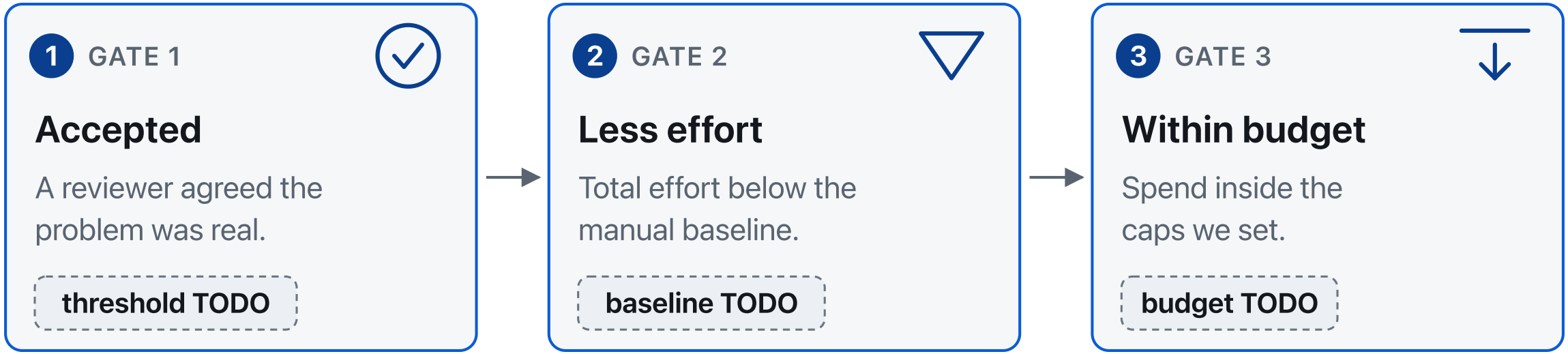


> Record accepted changes, reviewer effort and total cost together.

TODO · expected pull-request volume · a hard stop rule (for example, pause after N rejected drafts) · a setup-time estimate.

Fig. 6 · Week 1 runs with `staged: true`: writes are skipped, the run still costs.

Week four ends in a decision we agreed in advance



All three must pass; a miss is a real decision. Agreed before we start, so the result can tell us to stop.

TODO · all three thresholds, and the manual baseline they are measured against, need agreeing before week one.

Proposed criteria · this deck's recommendation, not an agreed plan.

The downside is bounded by five proposed limits

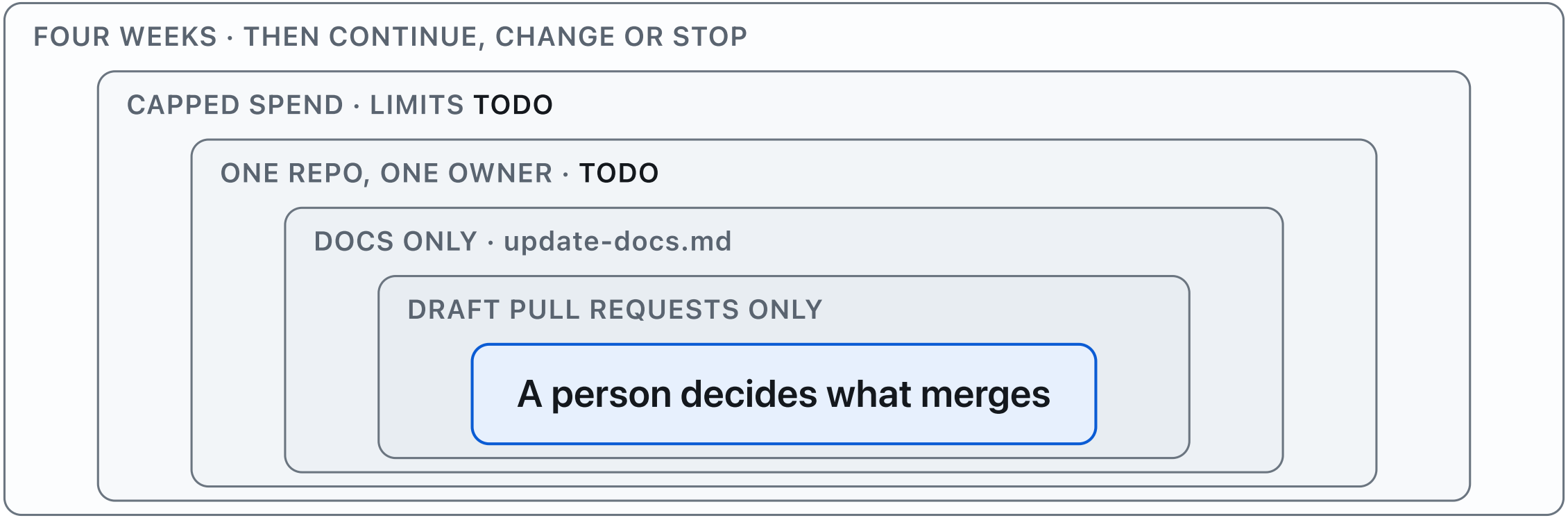


Fig. 7 · Proposed, not yet configured · spend, repository, owner TODO · "docs only" is by instruction and diff review (`protected-files` guards sensitive files, not paths) · a person deciding is our pilot policy, not a product guarantee · TODO branch-protection or required-review setting that enforces it.

PR policy (draft: true)

Approve a four-week, docs-only pilot

Public preview since **11 Jun 2026**: features and defaults can change.

PROPOSED STARTING POINT

Candidate, a public third-party template, not our repository: `githubnext/agentics workflows/update-docs.md`

Every change is a draft pull request; a person decides what merges. Switch off at any time.

TO DECIDE BEFORE WEEK ONE

- [ ] Pilot repository + owner TODO
- [ ] Per-run + total pilot budget TODO
- [ ] Week-four thresholds + baseline TODO

Reversible by design · pause on unsafe output or a breached limit · proposal, not yet approved.

APPENDIX 1 / COMPILATION

Compilation turns that file into an Actions workflow

1 · Author

Markdown task
and frontmatter

2 · Compile

`gh aw compile`
Generated `.lock.yml`

3 · Publish

Commit both files
Run in GitHub Actions

Compilation validates configuration and pins Actions to commit SHAs.

Agents add a third lane beside CI and CD

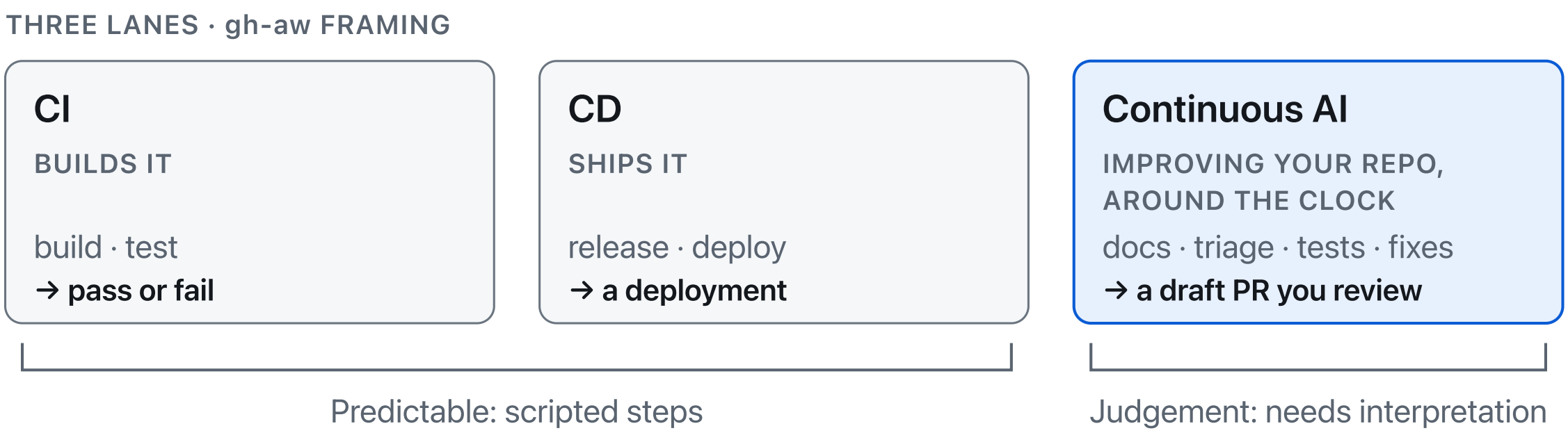


Fig. 8 · Lane captions quoted from the gh-aw docs · the predictable/judgement split is this deck's framing.

gh-aw docs: "A third pillar beside CI and CD"

Detection adds a review gate, not a safety guarantee

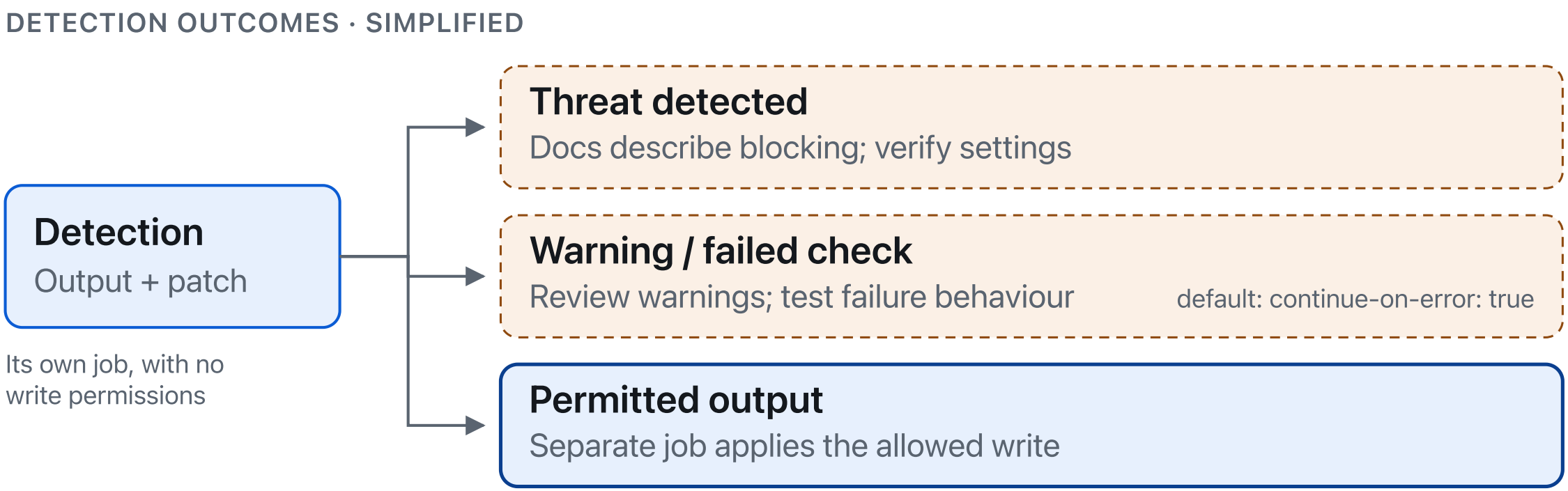


Fig. 9 · The threat-detection page describes both caution-only and blocking outcomes when a check fails; test explicit settings before relying on either.

APPENDIX 4 / DETAILS FOR QUESTIONS

Use agents where interpretation adds value

Good pilot candidate

Compare code with documentation.

Propose a small, reviewable correction.

Keep predictable work in CI

Build, test and release steps.

Use deterministic tools when they solve the task.

TODO · Agree which tasks stay with a person and our current autonomy level.

Start with one workflow from the example collection

Example collection

> TODO: repository URL
> TODO: workflow file
> TODO: recommended first task

Review before adding

Instructions and scope
Permissions and network
Outputs and budget

```
gh aw add-wizard owner/repo/workflow
```

Template, not runnable · replace it with the confirmed workflow reference.

A trial can run in a temporary private repository

Select

Choose the reviewed workflow file.

Trial

`gh aw trial <workflow>`
Private host repo by default; delete it after.

Inspect

Read the results before choosing the next step.

TrialOps is experimental. Isolation changes where the workflow runs; it does not make the run free or risk-free.

Staged mode previews safe outputs before they are applied

```
safe-outputs:
  staged: true
  create-pull-request:
    draft: true
```

Staged safe-output writes are skipped; the workflow still runs.

Compile and publish
Commit the Markdown and lock file.

Run and inspect
Read the Actions step summary.

Then decide
Enable draft PRs only after review.

The agent drafts; a person decides what merges

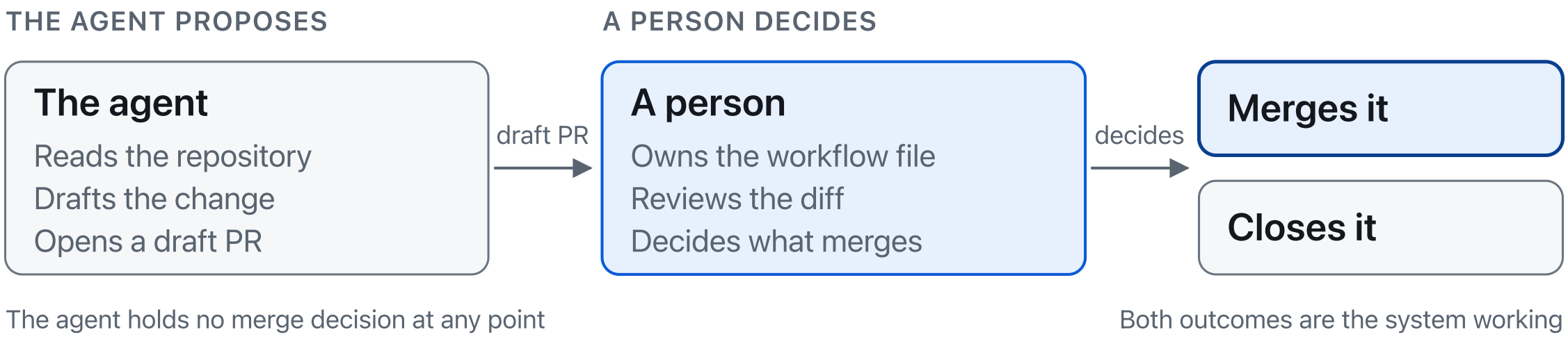


Fig. 10 · This deck's framing · `draft: true` is enforced pull-request creation policy · reviewer minutes are not yet measured.

Cost per accepted change is a formula the pilot fills in, not a number we have

$$C_{acc} = \frac{p_a A + p_m M + wR}{N_{acc}}$$

$$C_{man} = \frac{wR_{man}}{N_{man}}$$

Continue only if **$C_{acc} < C_{man}$**. The manual baseline is TODO.

Symbol	Meaning	Value
A	AI credits used (`gh aw logs`)	NOT MEASURED
p_a	price per credit	≈ $0.01, estimated
M	Actions minutes (billing)	NOT MEASURED
p_m	price per Actions minute	TODO
R	reviewer minutes	NOT MEASURED
w	reviewer cost per minute	TODO
N_{acc}	proposals merged	NOT MEASURED

This deck's formulation · no run yet, so nothing here is measured · AI credit (AIC) accounting is best-effort and billing may differ.

APPENDIX 10 / DETAILS FOR QUESTIONS

The technical claims are grounded in GitHub's documentation

Public preview since **11 Jun 2026**

Product and controls

Official changelog
Architecture and threat detection
PR policy and staged mode

Operating and measuring

CLI and schedule syntax
Cost management
Reported documentation results

Unresolved conflict · whether a failed detection run blocks safe outputs

Technical fact check: 17 Oct 2026; re-checked 19 Oct 2026. Sources are linked on the relevant slides and in speaker notes.

Public-preview changelog · Official docs